MARCH 2026

EMPLOYEE ATTRITION PREDICTION USING MACHINE LEARNING

RUNNING MACHINE LEARNING MODELS AND ROC CURVE ANALYSIS

Prepared by: Shashank G

Models Evaluated: Logistic Regression | Decision Tree | Random Forest

Tools Used: Python | Scikit-Learn | Pandas | Seaborn

02

INTRODUCTION

In this component, multiple machine learning models were implemented to predict employee attrition based on HR analytics data. The goal was to compare different classification algorithms and evaluate their performance using accuracy, confusion matrix, classification report, and ROC curves.

The models implemented were:

- Logistic Regression
- Decision Tree
- Random Forest

Each model was trained using the prepared dataset and evaluated on test data.

03

MACHINE LEARNING MODELS USED

LOGISTIC REGRESSION

Logistic Regression is a classification algorithm used to predict binary outcomes such as employee attrition (Yes/No). It works by estimating the probability that an employee belongs to a particular class.

Results: Accuracy: 87.75%

Key observations:

- Model shows strong overall performance
- Good detection of non-attrition employees
- Better balance between precision and recall
- Performs well as a baseline classification model

DECISION TREE

Decision Tree is a supervised learning algorithm that splits data into branches based on important features. It helps in identifying relationships between features and target variables.

Results: Accuracy: 81.97%

Key observations:

- Model captures feature relationships
- Performance lower than Logistic Regression
- Weak detection of attrition cases
- Some overfitting tendency observed

04

RANDOM FOREST

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.
Results: Accuracy: 87.75%
Key observations:

- Stable performance compared to Decision Tree
- Good prediction of majority class
- Lower recall for attrition class
- Accuracy similar to Logistic Regression

MODEL COMPARISON

Model	Accuracy
Logistic Regression	87.75%
Decision Tree	81.97%
Random Forest	87.75%

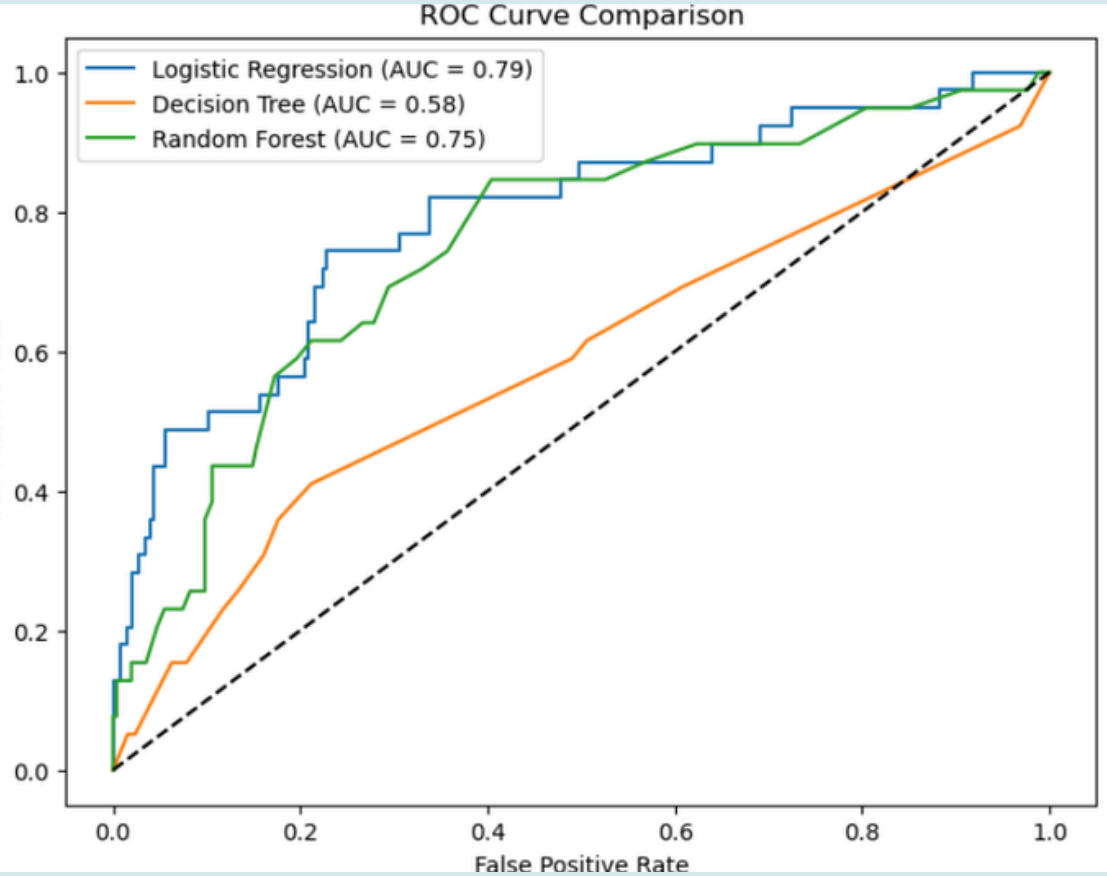
Observation:
Logistic Regression and Random Forest achieved similar accuracy, while Decision Tree performed slightly lower.
However, accuracy alone is not sufficient due to class imbalance.

05

ROC CURVE ANALYSIS

Receiver Operating Characteristic (ROC) curve is used to evaluate classification performance by comparing True Positive Rate and False Positive Rate.
Higher ROC score indicates better model performance.

Model	ROC Score
Logistic Regression	0.79
Decision Tree	0.58
Random Forest	0.75



ROC CURVE OBSERVATIONS

From ROC analysis:

- Logistic Regression shows highest ROC score
- Random Forest shows good performance but lower than Logistic Regression
- Decision Tree shows weakest classification ability
- Logistic Regression curve is closest to top-left corner indicating better performance

CONFUSION MATRIX INSIGHTS

Model comparison based on confusion matrices shows:

Logistic Regression:

- Better detection of attrition employees
- Balanced predictions

Decision Tree:

- Lower detection of attrition cases
- Higher misclassification

Random Forest:

- Strong prediction of non-attrition
- Weak prediction of attrition cases

07

KEY LEARNING FROM MODEL COMPARISON

The comparison shows that different models behave differently when predicting minority classes.
Important learnings include:

- Accuracy alone cannot determine best model
- ROC score provides better comparison
- Class imbalance affects model performance
- Logistic Regression shows most balanced performance

CONCLUSION

In this component, three machine learning models were implemented and evaluated to predict employee attrition.

Logistic Regression and Random Forest showed higher accuracy compared to Decision Tree. ROC curve analysis showed Logistic Regression achieved the highest ROC score, indicating better classification capability.

This component demonstrated the importance of comparing multiple models before selecting the final prediction model.

The final model decision will be made in the next component based on these performance comparisons.